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School Nutrition Policies and Practices on an American Indian Reservation: Research and Policy Considerations

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Abstract

Navajo (Diné) and other American Indian children are more affected by overweight and obesity than their U.S. counterparts. In this descriptive study, the authors combined a socioecological and Navajo cultural framework to analyze the various factors that influence food choices available to children in schools. The Healthy Hunger-Free Kids Act of 2010 updated nutrition standards for school meals and all other foods and beverages sold in schools. Some studies have shown that improved nutrition standards were not favorably received by students and school food programs. To better understand these broader trends in a Navajo context, the researcher administered in-person surveys to 6 principals and 14 food service workers from a sample of 6 schools and performed on-site observations of two schools. Data were analyzed using a combination of descriptive statistics, matrix analysis, and thematic analysis techniques. Results showed that schools were offering more healthy food options in school lunches and that some schools were still serving unhealthy foods, such as pizza, tater tots, and French fries. Classroom and fundraising events provided other sources for low-nutrient foods and beverages. Participants had mixed responses about the updated nutrition standards, and food waste of healthy foods was a major concern. This study proposes opportunities for nurses to engage in research and advance stronger policies that increase healthy food options and limit access to less healthy foods.

Keywords

American Indians, child, food services, overweight, obesity, surveys

Obesity disparities that begin earlier in life are a serious problem in Navajo (Diné) and other American Indian (AI) communities. While obesity estimates in AI children are not systematically collected and difficult to find (Wann et al., 2015), a 2017 obesity study in AI and Alaska Native children is perhaps the most recent and extensive study to date that indicates 45% to 51% of children aged 2 to 19 years are either overweight or obese (Bullock et al., 2017), whereas nationally, 17% of children and youth in the same age group are obese (Ogden et al., 2016). Alarmingly, there are also a few studies published on effective obesity prevention and treatment interventions for AI children (Wann et al., 2015). With growing evidence that have deemed schools as key environments for promoting and teaching healthy behaviors in children (Lavelle et al., 2012; Sobal-Goldber et al., 2013), this article addresses this gap and reports on how schools is one of the largest AI reservation in the United States, the Navajo Nation, have been meeting the nutritional needs for Navajo children.

The Navajo Nation covers more than 27,000 square miles in Arizona, New Mexico, and Utah and has more than 330,000 tribal members, with 40% of the total population represented by children and youth aged 19 years and younger (U.S. Census Bureau, 2010). The contemporary health of Navajo children is likely a result of

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generational changes in the cultural lifeways of the Navajo people. Changes in the cultural lifeways of the Navajo people from generations ago have negatively affected their nutritional health. The diet of Navajo people that once depended on food sources from farming, gathering, and hunting has nearly faded, eliminating a lifestyle that protected the health of the Navajo tribe. Many Navajo children not only have never experienced or witnessed this ancient lifestyle but also have never been taught sacred Navajo cultural teachings and practices (Begay, 2007; Kahn-John, 2010). Without foundational teachings, such as the Navajo concept known as *hózhó*, children are unaware of cultural principles that teach the proper ways of living for healthy development of well-being (Kahn-John, 2010).

Present and persistent conditions that continue to challenge efforts to improve nutrition include high poverty levels in AI communities, high unemployment, and low educational attainment rates (Office of Minority Health, 2020). Furthermore, access to affordable and quality foods, such as fresh fruits and vegetables, is limited in reservation communities. Some larger communities have grocery stores, and other smaller reservation communities may only have a convenience store or a gas station outlet that vends primarily low-nutrient, inexpensive foods. Families that desire low-cost, healthy foods may have to travel long distances to access the nearest grocery store (Lombard et al., 2014; Pardilla et al., 2013). Altogether, these contexts create significant challenges to accessing adequate and quality nutrition for Navajo children making schools a vital source of nutrition in reservation communities.

School Food Environment

Schools have become a focus for childhood obesity prevention efforts, because the school is an institution that fosters learning and development in children; given that children are in school for an extended time and may consume more than half of their daily intake there, the school can also be a setting for encouraging healthy eating behaviors (Welker et al., 2016). Schools are considered highly complex food environments because of the different types of food and beverage choices available to children—not only the National School Lunch Program (NSLP) and School Breakfast Program (SBP) but also because of "competitive foods," which are all other foods and beverages sold or offered through vending machines, á-la-carte cafeteria lines, school fundraising activities, classroom parties, and outside foods brought by students and parents (Welker et al., 2016).

Schools that participate in the NSLP and SBP are required to meet federal nutrition guidelines, and by complying with requirements, schools are eligible to receive cash subsidies for each meal served. Administration of federal school meal programs is divided between the U. S. Department of Agriculture (USDA, 2019) and state and local school food authorities. For AI tribes, they follow the same nutrition rules set forth by the USDA for all participating NSLP schools. While tribes have expressed interest to exercise their sovereignty and administer and manage their own school nutrition programs within the framework of P.L. 93-638 (Indian Self-Determination and Education Assistance Act), these programs remain with federal and state agencies. Consequently, this limits tribes in having the ability to manage the nutritional quality of foods served to children, including having less flexibility in offering various cultural foods (Garasky et al., 2016).

School nutrition standards were last updated by the Healthy Hunger-Free Kids Act of 2010 (HHFKA) and implemented at the start of the 2012 to 2013 school year. The HHFKA instituted major changes to improve school food environments by aligning school meals with the latest evidence-based nutrition guidelines and, for the first time, requiring competitive foods to follow healthier nutrition standards. Schools were required to offer more whole grains, fresh fruits, vegetables, and fat-free milk; reduce the sodium content of meals; and limit trans-fat content (National School Lunch Program and School Breakfast Program, 2013). Since the enactment of HHFKA, further legislation pertaining to reauthorizing child nutrition programs, which may affect school nutrition standards, has not been completed. A final rule that became effective February 2019 allows school nutrition programs flexibility regarding milk, whole-grain, and low-sodium requirements (USDA, 2018).

U.S. School Reactions to HHFKA

Since the updated nutrition standards went into effect in the 2012 to 2013 school year, there have been mixed reports about how U.S. elementary and middle schools adjusted to these new requirements. Some evidence suggests that many school districts and school food authorities initially grappled with pushback from students, with students opting out of meal programs, because of their unwillingness to accept healthier foods (Byker et al., 2014; Cohen et al., 2014). Other reported challenges included schools having to deal with increases in food waste, procurement of whole-grain foods, and difficulties meeting low-sodium requirements (Asada et al., 2017). Despite these initial challenges to implementation, a nationwide survey showed that although elementary students initially complained about school meals, a majority of the students (70%) liked the meals, and participation in school meal programs has returned back to pre-implementation levels (Turner & Chaloupka, 2014). These studies shed considerable light on nationwide reactions of students and school food services to

HHFKA, yet comparable descriptive data on school food environments for AI schools in general, and for the Navajo Nation in particular, are lacking.

AI Schools

The few AI studies relevant to school food environments that exist date as far back to the 1990s with the Pathways study, and other studies, as Pathways that were either specific to a single AI tribe or region and have unknown transferability to other contexts (Davis et al., 1999; Saksvig et al., 2005; Story et al, 2012). The Pathways study is well-known for being the largest and earliest (1990s) school-based randomized intervention trial that aimed to reduce obesity in AI children. The researchers enrolled participants from 41 schools across seven different tribal nation communities, including schools on the Navajo Nation (Davis et al., 1999). Notably, the study was the first to document that AI children were consuming high-fat breakfasts and lunches, drinking high-fat milk, eating large portions, and being allowed second servings of unhealthy foods in schools (Gittelsohn et al., 1999). While the overall goal of the study to reducing body fat in AI children was not attained through interventions in food services, physical activity, classroom curriculum, and family participation, secondary goals to reduce the fat content of school meals and increase health-related knowledge, attitudes, and behaviors were attained (Caballero et al., 2003).

Learning from Pathways, a smaller-scale, randomized obesity prevention study known as Bright Start was undertaken from 2005 to 2006 on a single Lakota reservation in South Dakota (Story et al., 2012). This study also reported that the nutrient content of school meals far exceeded the recommended federal nutrition standards (Story et al., 2012). Distinct from Pathways, Bright Start focused on younger children and implemented interventions to limit low-nutrient snacks in the classroom, offer more fruits, vegetables, and low-fat milk, increase family involvement, and provide health education regarding substituting water for sugar-sweetened beverages. Results showed no significant intervention-related effects on children's body mass index and body fat; however, the interventions resulted in the successful reduction of the fat content of school meals (Story et al., 2012). Although the researchers successfully implemented intervention strategies that improved the nutritional quality of school meals, the long-term impact and sustainability of these approaches in schools and their food service programs on obesity remain unclear. Comprehensive, up-to-date, AI-focused studies that periodically examine all aspects of the school food environment, including foods and beverages that AI children are accessing in and around schools, are still needed.

Socioecological Model and the Navajo (Diné) Concept of Hózhó

In this study, the researcher combined the conceptual framework of a socioecological model (SEM) with a Navajo cultural perspective, an approach that an increasing number of indigenous researchers are using to generate new knowledge by exploring the interface between an indigenous way of knowing and Western theory (Durie, 2004; Kovach, 2010; Lavallee, 2009). Townsend and Foster's (2011) SEM model was adapted and applied to understand diet choices and healthy eating in school settings and to guide the analysis and interpretation of influences on the complex food environments in this study. These levels of influences are depicted as five concentric circles, with students represented in the innermost sphere, encircled by local influences within schools that extend outward to more distant and broader influences outside of schools (Figure 1). Importantly, these ecological levels of influence symbolize the connections that Navajo children relate to within and outside of schools that ultimately influence their nutrition and health behaviors in schools. This view parallels a traditional Navajo perspective, specifically through the concept of *hózhó*. Although there are no words or phrases that can adequately capture the meaning of *hózhó*, it has generally been translated as being in harmony, balance, and wellness (Begay, 2007; Benally, 1987). In the context of this study, *hózhó* is a fundamental Navajo concept relevant to the health and well-being of Navajo children, and by understanding the different contextual elements relative to the school food environment, schools can be better positioned to create more

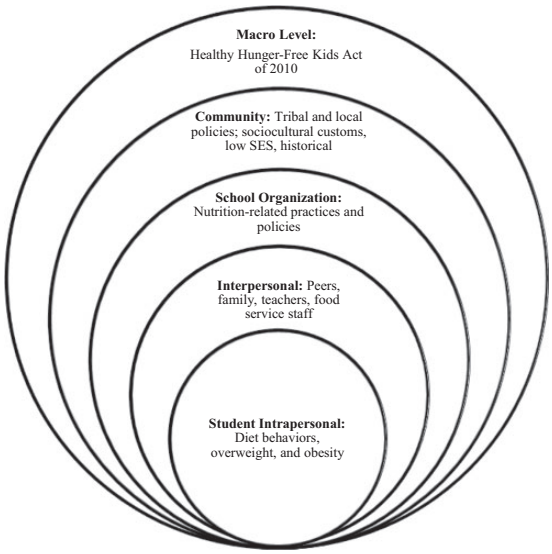


Figure 1. Adapted Version of the Socioecological Model.
Note. SES = socioeconomic status.
Source: Townsend & Foster (2011).

healthful food environments. Schools with healthy food environments, in turn, can promote healthy nutrition behaviors among Navajo children to attain and maintain optimal well-being, which is congruent with the precepts of *hózhó*.

Methods

A descriptive research design that employed both quantitative and qualitative methodologies was used to capture greater detail about the existing school food environment of select schools on the Navajo reservation. Descriptive research is useful for finding out the "what is" about a situation, subject, behavior, or phenomenon (Langford & Young, 2013). A combination of survey and observational methods was used. Given the scarcity of literature on this topic, a descriptive design was appropriate for describing the existing school food policy and practice environment.

Procedure

Research approvals were obtained from the University of New Mexico's Human Protections Office, participating Navajo school boards and communities, and the Navajo Nation Human Research Review Board. Using a convenience sampling method, the researcher selected six elementary and middle schools participating in the NSLP within the Navajo Nation. Given the vast geographic area of the Navajo Nation, only schools within a select region were invited to participate. In November and December 2016, 6 principals and 14 food service workers were administered an existing two-part survey that was modified for this study (Turner, Sandoval, et al., 2014). Permission to adapt the survey was obtained from the survey authors, changes included removing sections pertaining to physical activity and adding four open-ended questions to the end of the survey. The open-ended questions were aimed at gaining a better understanding of (a) experiences with implementation of the NSLP nutrition requirements, including barriers and facilitators, (b) Navajo cultural practices related to nutrition and health, and (c) the role schools play in nutrition for students. Principals completed Part 1 of the survey, which contained questions on general school characteristics, school food policy and practices, and school wellness. Food service workers completed Part 2, which consisted of 26 survey items that asked about the types of foods and beverages served in school lunches and competitive venues.

Because of the long distance traveled to reach the schools, which were located in remote reservation communities, participants completed the surveys at the time of distribution. The researcher explained the study and offered participants the option of having the surveys read to them and their responses recorded if they were not fully comfortable with self-completing the survey. Because none of the participants requested that the surveys be read, the researcher exited the room but remained close by in case questions arose. On submitting the surveys, a number of participants *voluntarily* shared more in-depth information (no planned interviews were conducted) related to the survey questions and issues of importance about school nutrition and food service programs. The researcher documented important points and key concerns from these conversations. This method of engagement and dialogue is consistent with indigenous ways of sharing knowledge through orality and storytelling traditions (Kovach, 2010).

The researcher conducted observations of two schools during same period of survey completion, mainly of the school cafeterias, hallways, and areas near or around offices before, during, and after lunch periods, totaling approximately 3 hours. An observational checklist was used to record the availability of vending machines, types of snacks or beverages sold, location of vending machines, and all other alternative food and beverage sources (Gordon et al., 2007). In addition, field notes to supplement survey data and illuminate a more holistic understanding of the context of the school environment were documented (Patton, 2015). Although observations were conducted as unobtrusively as possible, one school employee approached the researcher, sharing concerns and perspectives about school nutrition and the health of children. Key phrases and comments from this dialogue were also documented.

Analysis

Categorical responses to survey questions were described using frequency and percentages. To analyze textual information gathered from open-ended survey questions and conversations with participants at schools, a matrix analysis was completed by the researcher, which involved classifying and organizing raw data into key themes through a cross-referencing exercise that produced a final matrix display of data grouped by principals, food service personnel, and the researcher's reflections (Averill, 2002). For observational data, frequency counts were used to describe the type, number, location, and times of day vending machines were in operation. Content analysis and thematic analysis were used to analyze field notes on the school food environment (Patton, 2015).

Results

Twenty respondents from a total of six elementary and middle schools located in several remote communities participated in the study. Five of the schools were part

of the Arizona public school system, and one was a Navajo-controlled community grant school. All were low-income schools, with 100% of children enrolled in free or reduced-priced meals. Student enrollment ranged from 160 to 472 students per school.

School Food Environment Policies and Practices

The principals' responses related to school food policies and practices are listed in Table 1. Most principals were familiar with their school's wellness policies, yet approximately half indicated that they did not have a designated person to ensure their implementation. Most principals were not as familiar with school meal nutrition standards. Furthermore, half of the principals indicated having an active wellness advisory council, and half did not respond; few principals had restrictions in place regarding fundraising products and teachers offering candy as rewards in classrooms. Health screening and nutrition education were also not frequent practices in schools.

Table 1. Principals' Descriptions of School Food Environment Policies and Practices ($n = 6$).

Description	*n* (%)
Wellness policy and other nutrition-related policy	
Familiarity with the wellness policy	4 (67)
School/school district has one or more designated persons for ensuring the wellness policy is implemented[a]	3 (100)
School/school district has health advisory council	2 (33)
Extent of familiarity with USDA nutrition standards	
Not at all	1 (16)
A little	3 (50)
Somewhat	2 (33)
Extent school practices align with nutrition standards	
Have already made changes	2 (33)
Don't know	3 (50)
Not applicable	1 (16)
Has policies regarding nutritional quality for fundraising[b]	2 (40)
Classroom practices	
Teachers allowed to use candy as a reward for good academic performance[c]	3 (75)
Teachers allowed to use candy as a reward for good behavior[c]	3 (75)
Classroom lessons involve candy (math using M&M candies)[c]	3 (75)
Nutrition education provided in classroom[c]	2 (40)
Other health-related practices	
School has a garden that students participate in	3 (50)
Participates in USDA Fruit and Vegetable Program[b]	2 (40)
Body mass index screening of students	
Never	4 (67)
3Annually	2 (33)

Note. USDA = U.S. Department of Agriculture.
[a]Missing data from three principals. [b]Missing data from one principal. [c]Missing data from two principals.

School Lunches

Table 2 lists food service participants' responses to the types of foods and beverages offered in school lunches and foods sold through á la carte cafeterias, school stores, and snack bars. Most food service respondents (FSRs; 85%–100%) indicated that more *healthy* foods

Table 2. Food Service Respondents' Perceptions of School Food Environment Policies and Practices ($n = 14$).

Response	*n* (%)
School food services program	
No registered dietitian at schools	14 (100)
Food service manager has food safety/nutrition certification	14 (100)
Food services managed by food service management company	14 (100)
Other food-related practices	
Participates in farm-to-school program	4 (29)
No vending machines	14 (100)
No á-la-carte beverages or foods	9 (64)
No school store or snack bars	12 (86)
Food and beverage items available at lunch	
100% fruit or vegetable juice	8 (57)
Nonfat skim white milk	6 (43)
Nonfat skim flavored milk	5 (36)
Low-fat 1% white milk	11(79)
Low-fat 1% flavored milk	9 (64)
Whole or 2% milk[a]	7 (64)
Low-fat baked goods	5 (36)
Low-fat salty snacks	4 (29)
Low-fat ice cream or frozen yogurt	7 (50)
Cookies, cakes, pastries not low in fat	3 (21)
Whole-grain crackers	12 (86)
Crackers not whole grain	4 (29)
Bread sticks, bagels, or other breads (whole grain)	14 (100)
Bread sticks, bagel, or other breads (not whole grain)	6 (43)
Whole-grain breads	14 (100)
French fries or tater tots offered	9 (64)
Cheese sticks (not low in fat)	5 (36)
Vegetables[b]	13 (100)
Fresh fruit	13 (93)
Salad bar	12 (86)
Healthier pizza (whole-wheat crust, low-fat cheese, toppings)	14 (100)
Regular pizza	7 (50)

[a]Missing three responses. [b]Missing one response.

were served as part of school lunches, including healthier pizza (whole-grain crust, low-fat cheese, fresh vegetables), fresh fruit, salad bars, vegetables, and whole grains since enactment of the HHFKA. Along with guidelines-friendly foods, half or more of the food service workers also reported that their schools served *unhealthy* foods that were not consistent with nutrition standards, such as French fries, tater tots, and regular pizza. For beverages, most FSRs (64%) indicated that higher fat milk (whole or 2%) was still being served, but encouragingly, 79% of FSRs reported serving low-fat 1% milk, and 43% reported serving nonfat milk.

Other School Food-Related Topics

Table 2 shows the responses of 14 FSRs who completed Part 2 of the modified survey. None of the food service managers were trained as registered dietitians. Schools used an outside vendor or food service management company to manage operations and implement their food services programs, and none of the schools had food-related vending machines. Furthermore, competitive food options, such as á la carte cafeterias and school stores or snack bars, were not available in schools according to most FSRs.

Thematic Findings

Barriers and Facilitators. Table 3 describes perceived barriers and facilitators to implementation of nutrition requirements. The major barriers identified by the open-ended questions were frustration with the newest nutrition standards and the perception that requirements were too strict and led to food waste. Additional barriers included concerns about the nutritional quality of lunches, the lack of school district support, meager understanding about the nutrition requirements, and inadequate knowledge of nutrition standards among food service personnel. One food service manager disclosed that the school changed food vendors with the intention of improving the quality of foods being served and reducing food waste: "We still see some food waste. The key is to be creative in making meals that are tasteful for children to eat."

Participants cited fewer facilitators compared with barriers. The main facilitators were staff training on nutrition requirements and increases in healthier food options.

Navajo Cultural Practices and Foods. Responses to the open-ended question about the integration of Navajo cultural practices, including food practices, remarkably showed that none of the reimbursable school lunch programs served Navajo cultural foods. Many FSRs explained that contracting with food service management companies restricted the serving of any type of cultural foods. Some commented that before outsourcing their food services programs, "school lunches were more nutritious because they were made from scratch" and "traditional foods like blue corn mush, squash, and stews were part of the menu." Hence, the present-day cultural food practices were only promoted and preserved as part of school-wide cultural heritage days or important school events. Because students were familiar with and enjoyed traditional foods, many participants thought schools should offer more of these foods.

Table 3. Perceived Barriers and Facilitators to Implementation of Nutrition Requirements Reported by Principals and Food Service Workers.

Facilitators	Barriers
Staff training	Lack of district support
"Food service staff attend trainings."	"District personnel don't understand the requirements we have to meet."
Availability of healthy foods	Lack of familiarity with nutrition standards
"Our kids are eating more whole grains and foods."	"Some of us are not familiar at all with NSLP nutrition requirements."
"We now have a salad bar which gives students more options to eat more vegetables."	Frustration with new nutrition standards
Display of more health education material	"New standards has changed the ways kids eat. A lot of the kids waste their food now."
"There are more posters displayed throughout the school and the cafeteria where students and staff can see every day."	"When we clean up after lunch, we find whole apples or pears in the trash cans."
	"Kids complain that the food has no taste. They don't like whole grains because it has no flavor and it is too dry."
	"When we serve hamburgers or hot dogs, kids only eat the meat, they throw away the buns."
	Smaller portion sizes
	"Students and even parents have complained about smaller portion sizes; one parent told us we were starving the kids."
	Quality of lunches has not changed
	"There is still a lot of heat-up foods served. It would be good to see more fresh foods prepared."

Note. NSLP = National School Lunch Program.

Observational Findings

The researcher observed two schools on separate days. The first was a kindergarten-to-eighth-grade school, with more than 150 enrolled students. The lunch consisted of whole-grain spaghetti, whole-grain bread sticks, canned peaches, steamed zucchini, and two choices of milk (nonfat chocolate milk and nonfat white milk). This school did not offer alternative food choices, such as á la carte items, and did not have a school store. According to the food service manager, by choice, their food services program only served one meal option, a strategy to limit other food choices. No vending machines were noted on the school campus. A school and community calendar was on display at the main entrance that listed after-school activities, which often coincided with club fundraisers. A school employee voluntarily shared that school fundraisers were held once a week by a student club, and popular items sold included soda, chips, candy, and pickles.

The second school was larger, with more than 400 students in Grades 4 to 6. A beverage vending machine containing water and juice was located at the main entrance. Food services offered more food choices, including the reimbursable meal and foods sold á la carte. The food choices offered were turkey and cheese subs, pepperoni and jalapeno pizza, cheeseburgers, orange chicken with rice, beef and bean burritos, and French fries. No school stores were noted. A food service worker stated that the school store was eliminated, because snacks were taken into the classroom and became a distraction.

Discussion

To our knowledge, this baseline descriptive study is one of the first to comprehensively examine and provide insight into school food environment policies, practices, and experiences since enactment of HHFKA, as well as integration of Navajo cultural practices, on one of the nation's largest AI reservations. The findings introduce potential nursing and policy research opportunities. Furthermore, this study was unique in its use of a combination of a Western ecological perspective and a Navajo cultural lens to analyze and understand how schools are contributing to the nutrition and health of Navajo children. From a Navajo cultural perspective, it is particularly important to reflect on the growing concern for the health of Navajo children. In today's modern technologic society, many youth are challenged by not being as culturally connected to their families, communities, and, ultimately, traditional teachings as their parents and ancestors were. The majority of youth today do not speak or understand their Diné language, are unable to communicate with their grandparents, are uncertain of the clan identities of their mother, father, and maternal and paternal grandparents, and even do not practice the sacred offering of white corn meal to the Holy People at early dawn. These are the basic teachings embedded in the Navajo concept of *hózhó*, a lifeway that teaches healthy living and promotes health and wellness, schools can be established as an environment that promotes, teaches, and models *hózhó*.

At the school organization level, it is encouraging that all NSLP school lunches are meeting the required nutrition requirements mandated by HHFKA, with more healthful food items, such as whole grains, fresh fruit, and vegetables, being served to students in Navajo reservation schools. At the same time, it is concerning that some schools are providing other food sources outside the federally sponsored NSLP lunches that largely consist of low-nutrient food items, such as regular high-fat pizza and French fries. These findings suggest that school lunch programs offer children a selection of foods that may or may not be healthy, which calls for additional work to enhance food options in school cafeterias. Research has shown that as long as nutrient-poor foods are a food option, these foods will be selected and consumed over higher-nutrient foods such as fruits and vegetables, when available (Chriqui et al., 2014).

Other survey study results about the school food environment yielded important information. For instance, most schools did not have an active wellness council, some principals were not familiar with nutrition standards, few schools had policies in place for fundraising and classroom parties, and teachers rewarded good behavior with food and snacks such as candy. Furthermore, respondents shared that students commonly brought in outside foods such as hot Cheetos, and at one school, an active student council sold unhealthy foods and snacks during school hours for fundraising. As a whole, this study highlights important findings for policy and practice, demonstrating that much work remains to be done to improve school foods, including the establishment of practices and policies that will encourage access to more healthy foods and limit low-nutrient foods. In addition, these findings may suggest that schools are struggling with federal wellness policy mandates, particularly with the nutrition sections of school wellness policies. Participating NSLP schools are required to develop nutrition policies for all foods and beverages offered and sold in schools, including cafeteria meals and foods outside of cafeterias, such as classroom parties, school celebrations, fundraising, and foods offered as rewards (Local School Wellness Policy, 2010). Only future inquiry can permit better understanding of how schools are developing, implementing, and enforcing policies that support healthy school environments.

Another school nutrition practice shown in the study was a low-participation rate in USDA programs, such as Farm-to-School, Fresh Fruit and Vegetable Program, and school gardens, which can enhance food environments by increasing access to fresh produce, fruits, and vegetables. These findings are noteworthy and warrant additional research to understand why schools are not seeking these additional opportunities. For example, farm-to-school programs are growing in many tribal communities and reconnecting children and youth with local and traditional foods, as well as their indigenous heritage, culture, and customs (USDA, 2016). For schools serving Navajo children, finding ways to engage more participation is imperative.

Based on findings from the qualitative open-ended survey questions and dialogue with participants, students' reactions to nutrition mandates were the main barriers to implementation of nutrition requirements. Food waste practices, particularly of healthy food items such as fresh fruits and vegetables, are consistent with the literature (Niaki et al., 2017; Smith & Cunningham-Sabo, 2013). Given the nutritional benefits, these foods offer and the limited access to fresh fruits and vegetable in rural, high-poverty reservation communities, future studies that objectively measure exactly what and how much students are eating and wasting, and explore why students are throwing away foods are warranted. In addition, urging more school food service programs on the Navajo reservation to incorporate taste-testing programs with students could be a vital strategy for introducing new, healthy foods. Inviting students to take part in deciding food choices in the cafeteria might be an incentive for them to be more open to trying and accepting healthier foods.

Consistent with the literature (Tabak & Moreland-Russell, 2015), one small study among food service directors identified lack of district support as a barrier to implementation of nutrition requirements. Although understanding the reasons underlying the barriers associated with the lack of district support was beyond the scope of this project, this finding is disconcerting because of the important role leadership plays. Evidence suggests that schools do better with implementation of school nutrition policies when there is support from school personnel (Masse et al., 2013). Langille and Rogers (2010) also stated that school and community leadership are crucial factors in whether and how health-related policies are implemented, which provides yet another reason to explore and find ways to strengthen collaboration among key stakeholders responsible for carrying out nutrition policies. In addition, findings that some schools are integrating traditional Navajo teachings and practices through classroom curriculum and offering occasional traditional food items is an area that deserves encouragement and further attention, specifically exploring the extent to which policies support these practices, including the integration of traditional and cultural foods into the school food services program.

Study Limitations

Given the unique conditions of the Navajo Nation, this study was intended to provide valuable insight into current school food environments, highlighting areas where schools serving Navajo students are succeeding and areas where improvement is needed. Although other communities may find transferable knowledge in this work, the results of this study are not intended to generalize to other populations. The sample size for this study was small, which was more feasible for a study that required long-distance travel to reach schools located in rural and remote AI reservation communities. This study relied on self-report surveys, and social desirability bias and lack of knowledge of content areas are additional factors that could have affected survey data (Creswell & Plano Clark, 2011). It is also important to note that many of the participants offered additional information after surveys were collected, which may have been because participants were more comfortable with conversation rather than completing surveys. From a cultural perspective, this study could have been strengthened by incorporating full interviews, but time and resource constraints were barriers.

Additional Research and Policy Considerations

In AI tribal nation communities, there is a significant need and opportunity for nurses to become more involved in advocating and promoting child health, school nutrition and wellness, and ultimately prevention of childhood obesity. It is important for nurses to learn and understand the landscape in AI communities, specifically the complex roles and interplay of federal, state, tribal, and local school policies that address child health in schools. Having such an understanding would better prepare nurses to actively engage with tribal leaders, school administrators, and school boards to educate and raise their awareness of health issues and identify areas needing further research and policy action (Ellenbecker & Edward, 2016).

Importantly, the Navajo Nation has exercised its sovereign right to develop key policies, moving its nation toward self-determination. The Navajo Sovereignty in Education Act, enacted by the 20th Navajo Nation Council in 2005, is one example of a policy relevant to child health (Navajo Nation Council, 2005). The main provisions of this law outlined explicit provisions for education and education standards but lacks specific guidance for child health in education systems (Navajo

Nation Council, 2005). Given the important role schools play in the health of children, nurses in Navajo and other AI communities can collaborate with key tribal partners to further develop and strengthen tribal and school health-related policies. For example, at the tribal government level, the Navajo Nation can declare a policy position on improving nutrition on school campuses by limiting access to unhealthy foods in three main venues: (a) school meal programs, (b) competitive foods sold outside of the USDA meals, and (c) other food sources, such as classrooms, fundraising efforts, and sporting events. Furthermore, this policy could include a statement that supports a traditional foods program in schools. One way to begin the policy formation process would be to bring the issue or problem forward to relevant stakeholders such as local Navajo Nation school boards, grassroots coalition advocacy groups, tribal-level Board of Education and school board associations, to elicit their support and begin a course of action that would include development of potential policy solutions, and finally, follow the Navajo Nation protocol for the legislative process (Navajo Nation Office of Legislative Services, 2020).

As a final point for nursing, this study identified a major gap in continuous and periodic data on school food environments and policies (Briefel et al., 2009), which could be met by the establishment of a monitoring and surveillance system maintained by tribal nations and drawing on nursing expertise. With such a system in place, periodic assessments of school nutrition and health-related policies and practices of all schools within their reservation boundaries could be performed, which would be an essential step for continuous enhancements and progress based on data analysis and interpretation.

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